

Rudenko Varvara

AI Strategy Lead | Head of ML
Implementation & Analytics

Telegram @Rudenko_Varvara

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scopus.com

scholar.google.com

github.com/Rudenshtok

Technical Skills

Domain: Reinforcement Learning, Recommendation Systems, A/B testing, stochastic optimization, MDPs, convex analysis, MCMC

Tools: Git, LaTeX, Linux, Docker, Postman, Jupyter Notebook, REST APIs, Redmine

Languages: Python, SQL

ML/RL/Math: NumPy, SciPy, Gymnasium, Stable Baselines3

English C1

French A2

Education

2026 – 2029

MIPT PhD

Department of Control and Applied Mathematics

2024 – 2026

MIPT Master's degree

Department of Control and Applied Mathematics

2019 – 2023

MIPT Bachelor's degree

Department of Control and Applied Mathematics

Summary

Head of ML Implementation with expertise in mathematical optimization and team leadership. I connect business strategy with technical execution: managing AI/ML projects from ideation to production, coordinating developers and stakeholders, and ensuring measurable business impact. Strong academic background allows me to evaluate technical solutions critically and guide engineering teams effectively. Recognized expert in stochastic optimization and RL with offers from top-tier quantitative funds (XTX Markets, 2026), seeking to leverage deep technical expertise to drive scalable AI solutions, optimize business processes, and lead high-performing engineering teams in a dynamic commercial environment.

Work experience

Head and Leading Specialist in Machine Learning

2025 – today

Joint Stock Company "Channel One"

1tv.ru

Team Leadership: Lead a cross-functional AI team (ML engineers, developers, content editors), managing the end-to-end lifecycle of AI products; conduct sprint planning, prioritize backlog, and mentor junior specialists.

ML Strategy & Delivery: Lead end-to-end implementation of AI-agent orchestrator for digital platform; coordinate with Product, Editorial, and DevOps to align technical solutions with business KPIs.

Business Impact: Automated content moderation and personalization workflows, reducing manual processing time by 75%.

Technical Oversight: Evaluate architecture proposals, select technology stack, and ensuring model quality, scalability, and robust A/B testing frameworks — without daily hands-on coding.

Stakeholder Management: Present ML roadmap and results to non-technical leadership; translate business requirements into technical specifications for engineering team.

Lecturer

2023 – today

Reinforcement Learning course MIPT University/ Math statistics course HSE University/ MIPT Olympiad school

Designed and delivered courses on Reinforcement Learning and Mathematical Statistics for 100+ students per semester. Developed educational materials and practical assignments focused on real-world applications of ML. Mentored students on research projects; improved ability to explain complex technical concepts to diverse audiences — a skill directly applicable to stakeholder communication.

Research Associate

2024 – 2025

Artificial Intelligence Research Institute (AIRI)

Airi.net

Working on writing articles and obtaining new results in the field of stochastic optimization for MDP, results published in ICLR 2026

Research Associate

2021 – 2025

Laboratory of mathematical methods of optimization

LabMMO.ru

2025 Multi-agent Reinforcement Learning system in double auction setting – Creation of a MARL user environment in which several agents trade with each other in a continuous double auction

Additional Education

2024

Artificial intelligence in climate sciences

Sirius University of Science and Technology

2022

Modern methods of optimization

Sirius University of Science and Technology in the program "Modern methods of information theory and optimization"

2021

Data Analysis in the Industry

Tinkoff, "Data Analysis in the Industry"

2021

Fundamentals of Reinforcement Learning

The Alberta Institute & Alberta Machine Intelligence Institute

2021

Stochastic algorithms and machine learning

Sirius University of Science and Technology in the program "Modern methods of information theory, optimization and management"

2020

Sampling, management and optimization

Sirius University of Science and Technology in the program "Modern methods of information theory, optimization and management"

Charitable activity

2022 – 2024

Teaching mathematics at a charity school for cancer-stricken children

Charity Fund "Gift of life"

Research Associate

International Laboratory of Stochastic Algorithms and Multidimensional Data Analysis

🖱 HDILab.ru

2023 Regret Minimization for Discounted MDPs by Variance-Reduced Q-learning – New adaptation of classical Q-learning algorithm and introduce Q-learning with Hoeffding bonuses and Bernstein Bonuses

2021 UVIP: Model-Free Approach to Evaluate Reinforcement Learning Algorithms – Work with KBRL, KBSF methods, testing algorithm for two Gaussians

2020 Importance Sampling and control variates – Improving estimates by adding control variates with zero variance

Publications

Asynchronous Policy Gradient Aggregation for Efficient Distributed Reinforcement Learning

🖱 ICLR.cc

ICLR 2026

Algorithm for Constrained Markov Decision Process with Linear Convergence

🖱 Arxiv.org

AISTATS 2023

Markov Decision processes and convex optimization

🖱 CRM.org

CRM 2023

Conferences & Talks

🖱 65th All-Russian Scientific Conference of MIPT 2023 "Markov Decision processes and convex optimization"

🖱 Fall into ML 2023 "ALGORITHM FOR CONSTRAINED MARKOV DECISION PROCESS WITH LINEAR CONVERGENCE"

Grants

- ▶ Recognized by international quantitative finance industry (offer from XTX Markets, 2026)
- ▶ Top 100 MIPT. Winner of the "Research Breakthrough" category.
- ▶ High Professional Potential Group (High School Economics University Personnel Reserve) Category "New researchers"
- ▶ Laureate of the scholarship for achievements in the field of numerical optimization methods MIPT
- ▶ Optimization and control methods in the complex robotic systems with many degrees of freedom in a dynamic environment Sber Allab